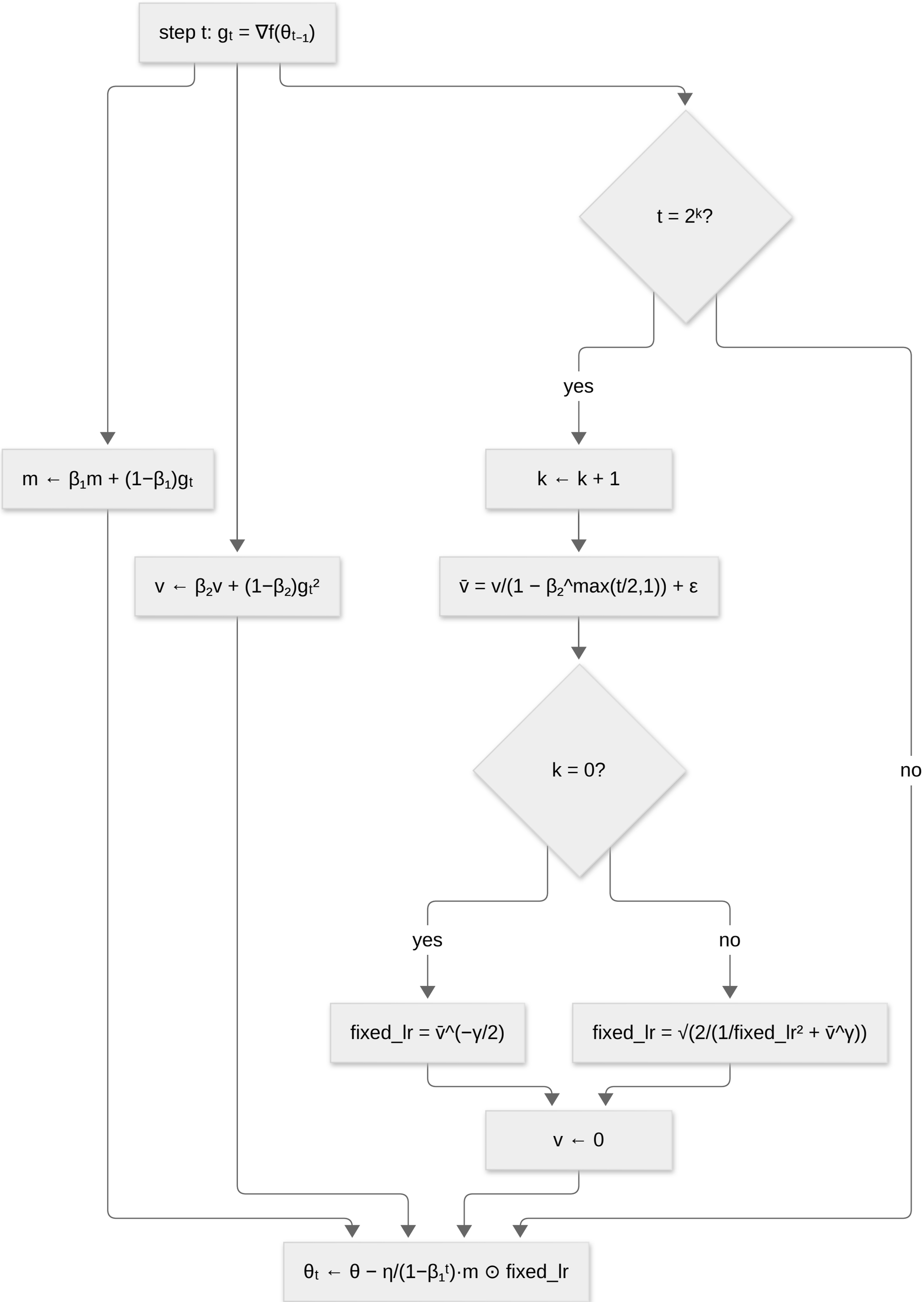

step t: $g_t = \nabla f(\theta_{t-1})$
$t = 2^k$?
yes
no
$m \leftarrow \beta_1 m + (1-\beta_1) g_t$
$k \leftarrow k + 1$
$v \leftarrow \beta_2 v + (1-\beta_2) g_t^2$
$\bar{v} = v/(1 - \beta_2\text{^max}(t/2,1)) + \varepsilon$
k = 0?
yes
no
fixed_lr = $\bar{v}$^(−γ/2)
fixed_lr = √(2/(1/fixed_lr² + $\bar{v}$^γ))
$v \leftarrow 0$
$\theta_t \leftarrow \theta - \eta/(1-\beta_1^t) \cdot m \odot$ fixed_lr